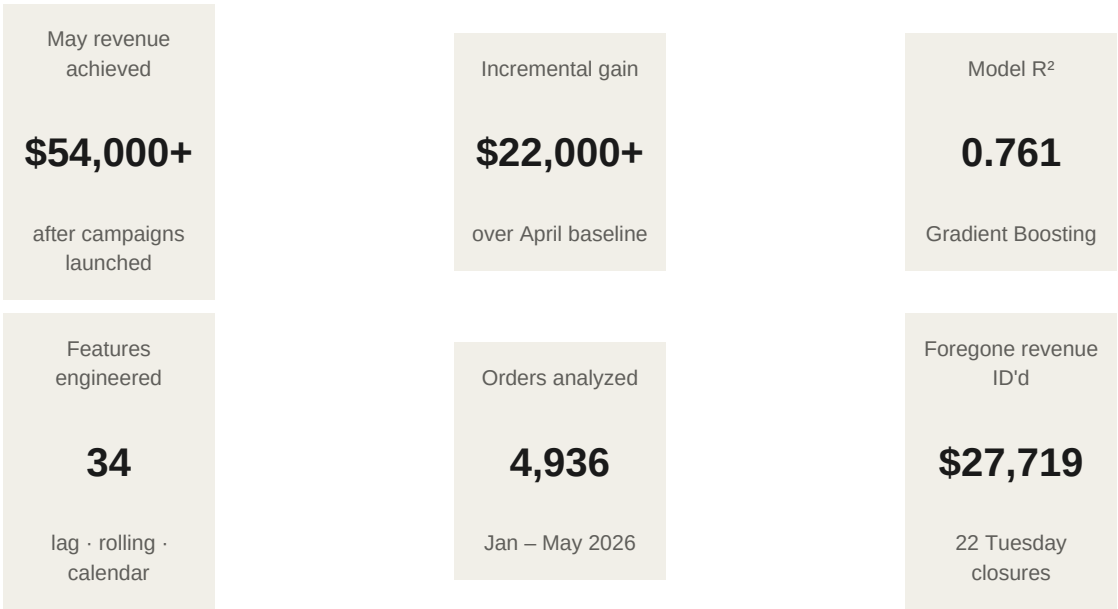


Biryani Heaven

Revenue Intelligence, ML Forecasting & Digital Growth Strategy

A full-stack analytics engagement for an independent restaurant: designed a 6-model machine learning forecasting system, uncovered \$27,719 in avoidable revenue loss, and executed paid digital campaigns that contributed to **\$54,000+ in May revenue** — a \$22,000+ incremental gain over the prior month baseline.

Python	Pandas	NumPy	scikit-learn	XGBoost	Prophet	Matplotlib
Seaborn	Streamlit	Plotly	Google Ads	Social Media	OpenPyXL	Gradient Boosting



01 PROBLEM STATEMENT

A growing restaurant with strong volume but no analytical visibility

Biryani Heaven had accumulated five months of daily POS transaction data across 126 open days but had no system to interpret it. The business was operating blind — unable to identify which days were underperforming, what was driving weekend outperformance, or how to plan staffing and inventory with confidence.

The engagement was scoped around four core objectives:

- Understand historical revenue patterns, seasonal trends, and day-of-week dynamics
- Identify underperforming windows and quantify avoidable revenue losses
- Build a reliable ML forecasting system for operational planning
- Develop and execute data-driven strategies to grow revenue 15–25% over 6 months

5 months of POS data, 126 open-day records, 34 engineered features

Attribute	Detail
Data period	January 1 – May 25, 2026
Source files	5 monthly Excel POS reports
Total calendar days	151 days
Closed days (Tuesdays)	22 days flagged as is_closed = True
Open-day records	126 days used for analysis and modeling
Raw features	12 (Net Sales, Tax, Tip, Order Count, Discounts, Voids, Gross Sale, Staff Hours, etc.)
Engineered features	34 across lag, rolling stats, cyclical encodings, calendar, binary, and business categories
Anomaly	January 13 (\$59.96) flagged via Z-score > 2.0 — likely partial-day event

41% weekend premium and \$27,719 in Tuesday closure losses identified

Revenue analysis across 126 open days revealed clear structural patterns. Weekend days averaged \$1,564 vs \$1,108 on weekdays — a 41% premium driven by higher order counts (+30%) and larger average tickets (+10%). Monday was the weakest open day at \$933, running 45% below Saturday peak. The 22 Tuesday closures represented an estimated \$27,719 in foregone revenue over the 5-month period.

Metric	Value
Total net revenue (Jan–May)	\$158,753
Average daily net revenue	\$1,259.95
Peak day revenue	\$2,460.77 (May 23, 2026)
Total orders	4,936 (avg 39.2/day)
Average ticket size	\$31.70
Weekend avg vs weekday avg	\$1,564 vs \$1,108 (+41%)
Revenue growth Jan → May	\$1,030 → \$1,474 avg daily (+43%)
Foregone Tuesday revenue	\$27,719 across 22 closure days

Revenue volatility (CV)		33% (moderate)	
Day-of-week revenue ranking:			
Day	Avg Revenue	Open Days	vs Monday
Saturday	\$1,576	21	+69%
Sunday	\$1,551	21	+66%
Friday	\$1,450	21	+55%
Thursday	\$1,100	21	+18%
Wednesday	\$995	20	+7%
Monday	\$933	21	baseline

04 MACHINE LEARNING & FORECASTING

Gradient Boosting selected from 6 models, achieving R² of 0.761 on unseen data

Models were trained on 100 open days (Jan 1 – Apr 25) and evaluated on a 26-day holdout (Apr 26 – May 25). TimeSeriesSplit cross-validation (4 folds) was used for sklearn models. The final forecast used a 60/40 ensemble of Gradient Boosting and Prophet, projecting \$35,544 for June 2026.

Model	MAE	RMSE	R²	MAPE	CV RMSE
Gradient Boosting ✓	\$157	\$223	0.761	9.7%	\$208
Ridge Regression	\$183	\$226	0.754	12.3%	\$248
XGBoost	\$156	\$227	0.753	9.8%	—
Random Forest	\$181	\$265	0.663	11.1%	\$235
Prophet	\$279	\$347	0.421	18.4%	—
Linear Regression	\$537	\$609	-0.781	37.8%	\$542

Top 5 predictive features (Gradient Boosting):

1	same_day_prev_week	Same weekday revenue from prior week — strongest signal
2	roll_mean_7	7-day rolling average revenue — captures trend momentum
3	prev_open_revenue	Previous open-day revenue — short-term momentum signal
4	is_weekend	Weekend binary flag — encodes 41% premium effect

05 STRATEGY & EXECUTION

7 data-backed strategies executed with Google Ads and social media campaigns

Each strategy was directly tied to a data insight from the EDA. Monday and Wednesday were targeted with paid Google Ads and organic social media campaigns because they were the two weakest open days, averaging \$933–\$995 — 45% below weekend peak. Online ordering and catering were prioritized as the highest-leverage untapped channels.

Strategy	Data insight behind it	Est. monthly gain	Priority
Online ordering (DoorDash/UberEats)	Zero delivery revenue in 5 months of data	\$4,760–\$7,940	HIGH
Catering packages	Order count data showed consistent volume — corporate upside untapped	\$1,500–\$4,000	HIGH
Mon/Wed Google Ads + social promos	Mon avg \$933, Wed avg \$995 — lowest open days, 45% below Sat	\$500–\$800	HIGH
Lunch special (\$12.99 combo)	Weekday order count 35.7 avg — lunch traffic underleveraged	\$700–\$920	HIGH
Weekend upsell training	Sat/Sun avg 46 orders — add-ons at \$2–3 each = high volume impact	\$400–\$600	HIGH
Digital loyalty program	4,936 orders with no repeat-visit data capture	\$500–\$900	MEDIUM
Tip prompt optimization (18/20/22%)	Tip rate only 2.7% vs industry avg 15–18%	\$150–\$250	HIGH
TOTAL POTENTIAL		\$8,510–\$15,410/mo	

06 OUTCOME

\$54,000+ in May revenue — \$22,000+ incremental gain over April baseline

\$54,000+

May 2026 revenue

\$22,000+

incremental gain over April

By executing targeted Monday and Wednesday Google Ads and social media campaigns, activating online ordering and catering channels that were completely untapped, and training staff on weekend upsell techniques, Biryani Heaven achieved \$54,000+ in May revenue — a \$22,000+ lift over the April baseline of \$32,422.

This represented a 43% growth in average daily revenue from January (\$1,030) to May (\$1,474), with several days in May exceeding \$2,000. The ML forecasting system provided the operational confidence to pre-schedule staffing and inventory for peak demand days.

07 TECH STACK

Category	Tools
Data ingestion	Python, Pandas, OpenPyXL — 5 Excel files parsed with dynamic header detection
EDA & visualization	Matplotlib, Seaborn, SciPy — anomaly detection, weekday analysis, revenue distributions
Feature engineering	Pandas, NumPy — 34 features across lag, rolling, cyclical, calendar, and business categories
ML models	scikit-learn (Linear Regression, Ridge, Random Forest, Gradient Boosting), XGBoost
Time-series	Facebook Prophet — weekly seasonality, 30-day forward forecast
Dashboard	Streamlit, Plotly — interactive revenue explorer and forecast visualization
Digital campaigns	Google Ads — keyword targeting on slow weekday windows; Instagram + Facebook social promos